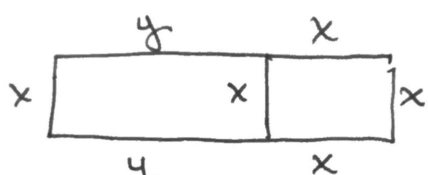
Course Pack Solutions - PAGE 8

1.



$$P = 5x + 2y$$

$$54 = 5x + 2y$$

$$y = \frac{54 - 5x}{2}$$

$$A = L \cdot W$$

$$= x\left(\frac{54 - 5x}{2} + x\right)$$

$$= x\left(27 - \frac{3}{2}x\right)$$

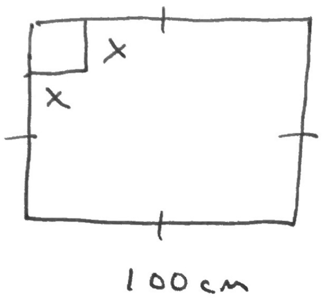
$$A' = 27 - 3x$$

$$A' = 0$$

$$27 - 3x = 0$$

$$x = 9 \Rightarrow y = \frac{9}{2}$$

2.



$h = x$

$L = 100 - 2x$

$w = 100 - 2x$

$$V = x(100-2x)^2$$

$$V' = (100-2x)^2 + x\left[2(100-2x)(-2)\right]$$

$$V' = 0$$

$$(100-2x)\left[(100-2x) - 4x\right] = 0$$

$$(100-2x)(100-6x) = 0$$

$\therefore x = 50$ or $x = \frac{50}{3}$

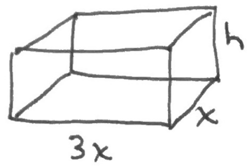
$\uparrow$

inadmissable

$\therefore h = \frac{50}{3}$ cm

$L = W = \frac{200}{3}$ cm

3.



$$A = (2x)(3x) + 2xh + 2(3xh)$$

$$200 = 6x^2 + 2xh + 6xh$$

$$200 = 6x^2 + 8xh$$

$$h = \frac{25}{x} - \frac{3x}{4}$$

$$V = x(3x) \cdot h$$

$$V = 3x^2\left(\frac{25}{x} - \frac{3x}{4}\right)$$

$$V = 75x - \frac{9x^3}{4}$$

$$V' = 75 - \frac{27}{4}x^2$$

$$V' = 0$$

$$75 - \frac{27}{4}x^2 = 0$$

$$x^2 = \frac{300}{27}$$

$$x \doteq 3.33\text{cm}$$

$$3x = 10\text{cm}.$$

$\therefore$ Base Dimensions are

$3.33\text{cm} \times 10\text{cm}$

4. $V = \pi r^2 h$

$S.A. = 2\pi r^2 + 2\pi r h$

$250 = \pi r^2 h$

$SA = 2\pi r^2 + \frac{500}{r}$

$h = \frac{250}{\pi r^2}$

$SA' = 4\pi r - \frac{500}{r^2}$

$SA' = 0$

$4\pi r - \frac{500}{r^2} = 0$

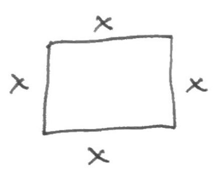
$4\pi r^3 - 500 = 0$

$r^3 = \frac{500}{4\pi}$

$r \approx 3.41$ cm

$h \approx 6.83$ cm

5.



$SA = x^2 + 4xh$

$3600 = x^2 + 4xh$

$h = \dfrac{3600 - x^2}{4x}$

$h = \dfrac{900}{x} - \dfrac{x}{4}$

$V = x^2 h$

$V = x^2\left(\dfrac{900}{x} - \dfrac{x}{4}\right)$

$V = 900x - \dfrac{x^3}{4}$

$V' = 900 - \dfrac{3x^2}{4}$

$V' = 0$

$900 - \dfrac{3x^2}{4} = 0$

$x^2 = 1200$

$x \doteq \pm 34.6$ cm

$\uparrow$

inadmissable

$\therefore x \doteq 34.6$ cm

$h = 17.3$ cm.

Volume $= 34.6 \times 34.6 \times 17.3$

Volume $\doteq 20{,}760$ cm^3

7 Done in class.

See class notes.